PSODNet: Pretrained Scene-Aware Object Detection for Optical Remote Sensing Imagery

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***Abstract*—With the widespread application of remote sensing (RS) images in military and civilian fields, RS object detection (RSOD) has become an important research direction. However, limited generalization and complex background interference have long been persistent challenges that hinder the development of RSOD. To address these issues, we propose a pretrained scene-aware object detection network (PSODNet). First, we design an enhanced object network (EON), which leverages a multihead pretraining strategy to jointly train data from diverse sources, thereby expanding the scale of dataset and improving the generalization ability. Second, we introduce scenario–object relationship module (SORM) to learn a multiscale relationship map between objects and scenes, which is used to constrain the solution space of object detection, thereby enhancing performance in complex scenarios. Finally, using label smoothing loss, PSODNet leverages mutual information of label to prevent extreme distributions of classification probabilities and reduce the risk of overfitting. In the experiment part, PSODNet was pretrained on multiple datasets and then fine-tuned on three datasets for validation. Results on three public datasets demonstrate that PSODNet outperforms existing models in detection performance by up to 2.4%, achieving a maximum mean average precision (mAP) of up to 95%. Through visual interpretation of the relationship map, we found that PSODNet is able to bridge the semantic relevance between objects and scenes, demonstrating its potential in object detection in complex scenarios. Code is available at: https://github.com/creature-compound/PSODNet**

***Index Terms*—Complex background interference, enhanced object network (EON), label smoothing loss, pretrained scene-aware object detection network (PSODNet), remote sensing object detection (RSOD), scenario–object relationship module (SORM).**

I. Introduction

With advancements in satellite technology, the resolution of remote sensing (RS) images has consistently improved, enabling clear depiction of various geospatial objects, such as ships (SHs), vehicles and aircraft [1], [2], [3]. These images not only capture more intricate details but also allow for a more precise representation of the objects' geometric structures [4]. As deep learning has progressed, RS object detection (RSOD) has rapidly gained traction as a prominent research area. Currently, RSOD is widely applied across both military and civilian sectors, including intelligent surveillance, urban planning, precision agriculture, and geographic information system (GIS) updates [5], [6], [7], [8].

The core task of RSOD is to detect, identify, and localize objects in RS images. Current object detection methods primarily rely on high-resolution optic images, using candidate box detection networks to perform object detection and recognition. However, as shown in Fig. 1, RS observations are often influenced by complex environmental factors [9], such as illumination variations and occlusion between objects, which result in a high false negative rate in detection models. In addition, RS images typically have a large swath width, making objects often small and with weak features, causing them to be easily obscured by complex backgrounds [10]. Besides, the geospatial distribution and arrangement of objects are often highly complex, increasing the challenge of distinguishing objects from the background. This results in challenge of variations in object scale [11]. Meanwhile, RS images also exhibit challenge of rotation variance due to varying sensor viewpoints and object orientations [12], [13], [14]. These factors contribute to detection models having limited adaptability and poor generalization in real-world environments.

The general training of object detection models is crucial for adapting to highly diverse real-world scenarios. The most direct way to enhance the generalization ability is to train on large number of parameters with large-scale datasets. For example, models such as ChatGPT [15], which are trained on millions of samples and have hundreds of billions of parameters, have been widely applied to various cross-domain tasks. In the field of RS, nearly all the object detection models are pretrained on the ImageNet-1K dataset, one of the most well-known image datasets in computer vision. The millions of real-world images from 1000 different categories enable these models to learn powerful representations. However, a significant domain shift exists between natural scenes and RS imagery, which makes ImageNet pretrained models (IMP) less effective at capturing the unique features of RS images [16].

In recent years, the number of satellites has increased significantly. According to a report by the Union of Concerned Scientists, over 1000 Earth observation satellites were in orbit, enabling the acquisition of extensive RS datasets and the training of large-scale RS models possible. For example, the Zhuhai-1 satellite generates nearly 20 TB of new RS

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data daily [17], [18], [19], [20]. However, annotating object bounding boxes for RS data is time-consuming and inefficient, resulting in a situation that the RS data are large but the annotated data are insufficient, which still hindered the training of large RS model. Some researches [21] found that annotating scene-level RS images is simpler than annotating bounding boxes and constructed the large-scale Million-AID dataset, which approximates to the volume of ImageNet-1K. Based on Million-AID, the large RS model can be achieved (RS Pretraining, RSP [22]), which is capable of representing general RS features. In addition, recent works have explored weakly supervised object detection in RS using scene-level annotations, which can not only avoid annotating bounding boxes but also improve model detection ability. Wang et al. [23] propose a novel multiview noisy learning framework, which solves the challenge of complex backgrounds and densely arranged objects. Wang et al. [24] propose a novel multiple instances learning paradigm based on negative deterministic information to solve discriminative instance domination and missing instances. Yao et al. [25] use a dynamic curriculum learning strategy to gain high-quality samples to improve model detection ability. However, a domain gap still exists between the weakly supervised pretraining tasks and the downstream object detection tasks, which limit the model's ability to effectively detect and localize RS objects.

Besides, with the rapid development of urban areas, artifacts have gradually increased, resulting in a significant increase in the heterogeneity of objects' distribution. To better handle these situations, researchers have focused on multiscale architecture or feature fusion strategy. For example, Wang et al. [26] introduced a single-stage detector called the full-scale object detection network (FSoD-Net), which consists of a multiscale enhanced backbone with cascaded scale-invariant regression layers. Wu et al. [27] designed a feature and spatial alignment network (FSANet) to improve the representation of small objects for more accurate localization. Dong et al. [28] developed a receptive field expansion block (RFEB) to enhance the detection accuracy of multiscale objects.

However, these approaches have focused primarily on the objects themselves, which overlooked the relationships between objects and the global scene. This oversight can result in missed objects in complex scenarios, where, for instance, recognizing road scenes improves vehicle detection under shadow occlusion. Furthermore, higher spatial resolution means that the difference between objects of the same category will be amplified, resulting in a larger intraclass variance and a relatively smaller interclass variance, which leads to a decrease in object recognition performance. Some methods focused on the perspective of loss function and amplify the interclass variance by adjusting the class weights, thereby making the intraclass variance relatively smaller, such as adaptive saliency bias loss (ASBL) proposed by Sun et al. [29], which adjusts category weights to improve object recognition, typically use hard labels. However, overemphasize interclass variance while neglecting intraclass variance would increase the risk of model overfitting.

To tackle the aforementioned issues, this article proposes pretrained scene-aware object detection network, a scene-aware object detection network designed to exploit the scenario–object relationship information in RS images, thus enhancing the generalization capability of object detectors across multiscale RS imagery. Three main contributions are summarized below.

1) To improve the model's generalization and mitigate the domain shift, we propose an enhanced object network (EON) to generate pretrained backbone specific for object detection task. During the pretraining stage, instead of pretraining on scene-level task in a traditional manner, we added the object detection decoder and replaced the scene classification head with detection head. At the same time, the limitations of current datasets, including insufficient sample sizes for pretraining and inconsistent classification systems across datasets, hinder effective joint training. Therefore, we propose a multihead pretraining strategy that integrates data from diverse datasets. This method expands the scale of dataset, enabling joint training, which in turn facilitates object-level RS pretraining.
2) In addition, we propose scenario–object relationship module (SORM), which leverages the patterned distribution of geospatial objects and the similarity of scenes where similar objects appear. By extracting scene embeddings from the backbone and object embeddings from the FPN, this module generates a scenario–object relationship map. This map uses scene knowledge to constrain the solution space of potential objects, enhancing object feature discrimination, reducing background interference, and improving detection accuracy in complex scenes.
3) To address the overfitting issue, we used label smoothing loss instead of traditional cross-entropy loss. Unlike conventional methods that focus solely on increasing interclass variance without accounting for intraclass variance, label smoothing incorporates a smoothing term to prevent extreme probability distributions, which effectively reduces intraclass variance, promoting robust generalization and mitigating overfitting.

The rest of this article is organized as follows. Section II briefly introduces the related work on RSOD and pretraining. Section III then describes the details of the proposed PSOD-Net. Section IV reports the comparative results obtained on three RSOD datasets and further discusses the specific role of the scenario–object relationship map. Finally, Section V concludes this article.

II. Related Work

With the rapid development of deep learning techniques, more and more object detection methods based on natural images have been proposed. However, due to the significant differences between RS images and natural images in terms of imaging features, object scale, and background complexity, the direct application of detection methods for natural images often leads to performance degradation. Therefore, to address the characteristics of RS images, researchers have gradually developed algorithms applicable to RS target detection. First,

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we review the object detection methods based on natural images and point out the limitations of these methods in RS images. Then, to address the special challenges including multiscale effects and background complexity, we present the proposed improvements to cope with these problems. Finally, when applying the model in the real world, there is a problem of insufficient generalization, and to address this problem, we discuss the application of pretrained models in the field of natural and RS images and propose solutions.

A. Natural Image Object Detector

With the rapid advancement of deep learning technology, numerous convolutional neural network (CNN)-based object detectors have been developed. Initially, research focused on anchor-based detectors, which are divided into one-stage and two-stage manner. For two-stage detectors, the most prominent example is Faster R-CNN [30], which consists of a region proposal network (RPN) and a region-based convolution neural networks (R-CNN) [31] for object detection. Subsequently, detectors such as Cascade R-CNN [32] and Trident-Net [33] were proposed to enhance performance. In contrast, one-stage detectors are advantageous in terms of ease of deployment and real-time performance, although their accuracy is relatively lower. These detectors directly predict class probabilities and bounding-box coordinates, with representative methods including SSD [34], RetinaNet [35], and YOLOv3 [36].

All the above methods require a large number of anchor boxes, which introduces significant hyperparameters and computational complexity. In addition, clustering anchor box sizes according to the dataset is a nontrivial process. To address these issues, anchor-free algorithms have been proposed in recent years, which can be categorized into keypoint-based and center-based detectors. Keypoint-based detectors first locate several predefined or self-learned keypoints, and then generate bounding boxes to detect objects. Examples include CornerNet [37], Grid R-CNN [38], and CenterNet [39]. CenterNet uses an hourglass network to predict the center, width, and height of the object, while CornerNet detects a pair of corner points of the bounding box and pairs them to form the final detection result. Center-based detectors define the center of the object as the foreground positive sample and then predict the distances from the center to the four edges of the bounding box to complete the detection. Typical methods include DenseBox [40], FCOS [41], and FoveaBox [42].

Due to the domain gap between natural images and RS images, object detection algorithms based on natural images are difficult to directly apply to RSOD.

B. RS Object Detector

RSODs have unique challenges such as the arbitrary direction of objects and the difficulty in detecting small and faint objects. And researchers designed special algorithms to address those challenges. For example, to adapt to the arbitrary direction of features, Xie et al. [43] proposed a general two-stage oriented detector called Oriented R-CNN, which can ensure high detection accuracy while taking into account the detection efficiency. Zhang et al. [44] proposed Cad-net, which captures the global and local context of an object by capturing the correlation and thus enhancing the recognition of objects. Considering the difficulty in detecting small and faint objects, Liu et al. [45] propose a new RS image tiny object detection network, which enhances the features for detecting tiny objects by dynamically adjusting the multiple receptive field features. Zhao et al. [46] used the relationship between contextual scene information and intrinsic features of dense tiny objects, suppressing the background information while enhancing the foreground information based on the scene information, to ensure accurate detection of tiny objects was realized.

In addition to the challenges mentioned above, in RS scenarios, due to the broad satellite viewing angle and the varied sizes of objects, the same type of object can exhibit a large-scale multiscale effect within a single image. Object detection algorithms designed for natural images are not well-suited to address this issue, leading to suboptimal detection performance.

To tackle the multiscale problem, researchers have proposed various approaches. Wang et al. [47] introduced the multicenter point learning network (M-CenterNet) to improve the localization performance of small object detection. Ma et al. [9] proposed a feature splitting and merging enhancement network aimed at emphasizing the features of small objects and suppressing background information. Liang et al. [48] proposed a single-shot small object detector (FS-SSD) based on feature fusion and scaling. In addition, Lu et al. [49] developed a precise and fast single-shot detector (AF-SSD) that integrates shallow and deep feature information through a multilayer structure. Singh and Davis [50] created an image pyramid to generate a set of images with different spatial resolutions, which serves as data augmentation for training with scale-awareness and as an ensemble method during inference to adapt to data distribution of multiscale object instances. In addition to the multiscale effect, challenges from complex backgrounds could also deviate the detection. To overcome this problem, Zheng et al. [51] proposed HyNet, which models the interaction between the foreground and the background to improve the model's performance. Zhang et al. [52] used an augmentation algorithm to adjust the weights of objects and backgrounds, expanding the distribution of training samples while reducing the interference of RS complex backgrounds on object features.

Inspired by the previous work, this study developed the global context fusion module (GCFM) to model the relationship between the object and scene, using global scene knowledge to constrain the potential scope of objects, thus improving the model's recognition ability in complex scenarios.

C. Pretraining for RS Detection

Pretraining has been demonstrated to have significant improvement on downstream object detection tasks for natural image. Typically, a self-supervised manner is adapted on a large-scale of nature images to train a robust backbone. After then, the trained backbone can be fine-tuned on downstream object detection task. SoCo [53] developed an instance-level multiscale contrastive learning network and

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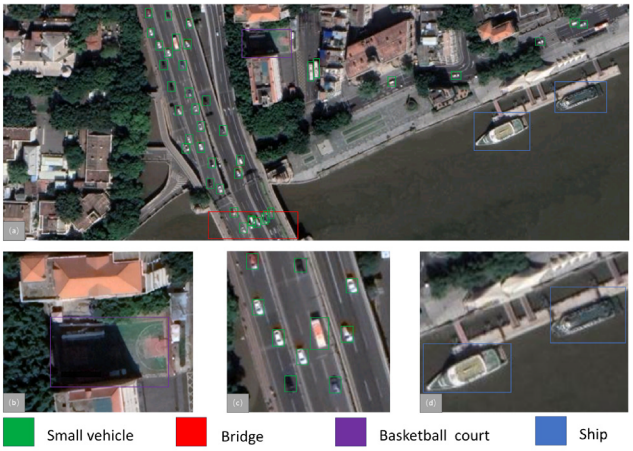

Fig. 1. (a) Typical RS image, with bounding-box annotations for multiple instances across various categories. (b)–(d) Zoomed-up region. (b) Occlusion of objects by shadows. (c) Significant scale difference between similar objects can be seen. Furthermore, high-resolution imagery amplifies intraclass variance among similar objects, as illustrated in (c).

trained it from scratch. However, as the parameter of the self-supervised learning (SSL) network is extremely large, it requires considerable computational resources. To mitigate this, later studies often freeze the pretrained backbone part, and only fine-tune the downstream-related part. This approach can save resources while retaining the backbone's generalization ability. Models such as UP-DETR [54] and DETReg [55] generate object embeddings using the pretrained backbone network and learn object representations by aligning with these embeddings. ProSeCo [56] suggests that discrepancies between object embeddings and detector features can impact pretraining effectiveness, thus proposing a teacher–student self-supervised pretraining method.

Given the domain gap between RS and natural images, multiple rounds of pretraining are a reasonable approach. Gururangan et al. [57] demonstrated that unsupervised pretraining on task-specific data can significantly improve performance in natural language processing (NLP) tasks. Inspired by this, Zhang et al. [58] developed a sequential pretraining strategy, first training on ImageNet (IMP [16]) and then pretraining on RS datasets with the masked image module (MIM) method. Similarly, Wang et al. [22] focused on scene recognition pretraining using the Million-AID dataset. Dery et al. [59] proposed a human-learning-inspired strategy, first performing contrastive SSL on natural images, freezing shallow layer weights, and then performing SSL on RS datasets. In addition to SSL, the task-aware training [59] can also be enhanced by integrating auxiliary data. For example, Wang et al. [60] used segmentation models such as UperNet [61] and Mask2Former [62] and introduced additional large-scale datasets for pretraining (SEP), thus improving performance in segmentation tasks.

Despite these advances, pretraining for object detection in RS images remains underexplored. To address this gap, we proposed an EON to improve object detection performance via pretraining. Following pretraining, the EON part is retained, enhancing the model's feature extraction ability for RS images during fine-tuning.

III. Methodology

The proposed PSODNet framework is divided into two stages: pretraining and fine-tuning. In the pretraining stage, the EON is used, while the fine-tuning stage consists of the SORM and the detection head, as shown in Fig. 2.

In EON, the object detection decoder and detection head are constructed on the traditional scene-level pretraining network whose scene classification head is replaced to mitigate the task-domain shift, and it is pretrained on multiple large-scale object detection datasets such as DIOR, HRRSD, LEVIR, and DOTA by multihead pretraining strategy.

During the fine-tuning stage, SORM extracts both object and scene information to learn scenario–object information. In SORM, instead of only generating the object embeddings, the feature embeddings of the whole scene are also established, which can additionally learn the scenario–object relationship map by the interaction of the two embeddings. The relationship map can then be used to improve the object discrimination with the constrain of scene information.

After getting the feature which is organized by the scenario–object relationship map, the detection head can use it to predict the anchor box and class. The detection head uses the two-stage design, comprising a proposal stage and a refinement stage. In the proposal stage, large-scale features are used to generate candidate object regions, each assigned a binary confidence score. The refinement stage then crops these candidate regions to produce refined bounding boxes and their corresponding multiclass scores. After multistage nonmaximum suppression (NMS) postprocessing, the final detection results can be obtained.

A. Enhance Object Network

In object detection, a well pretrained backbone can enhance the detector's performance. Many current RSOD methods leverage techniques from the natural image domain, such as using the pretrained backbone trained on large-volume ImageNet dataset. However, there is a significant domain gap between RS and natural images, leading to an intensive performance drop when using IMP models on RS images. To address this issue, subsequent studies have used the pretrained model trained on RS dataset, such as Million-AID, for RS task, which has million-scale volume similar to ImageNet-1K.

However, both IMP and RSP rely on scene-level tasks for pretraining, which has task-domain discrepancy with the downstream object detection task, and thus hinder the effectiveness of transferring pretrained models to downstream tasks. To bridge the gap between pretraining and downstream fine-tuning, we unified the task domain of two stages by the proposed EON. It additionally concatenates the object detection decoder with the traditional scene-level pretraining network and replaces the scene classification head with detection head. Besides, to address the limitations of current datasets, such as insufficient sample sizes and inconsistent classification systems that hinder joint training, we propose

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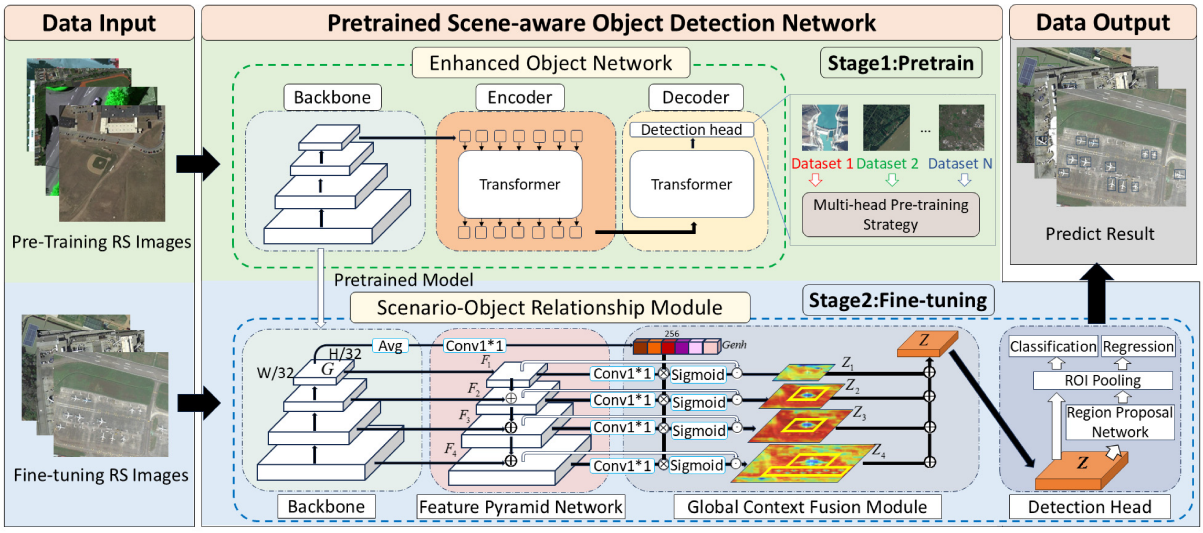

Fig. 2. Architecture of PSODNet. The pretrained backbone of the EON is used for SORM. SORM is composed of the backbone, FPN, and GCFM. The detection head is equipped with a detection head based on Faster R-CNN.

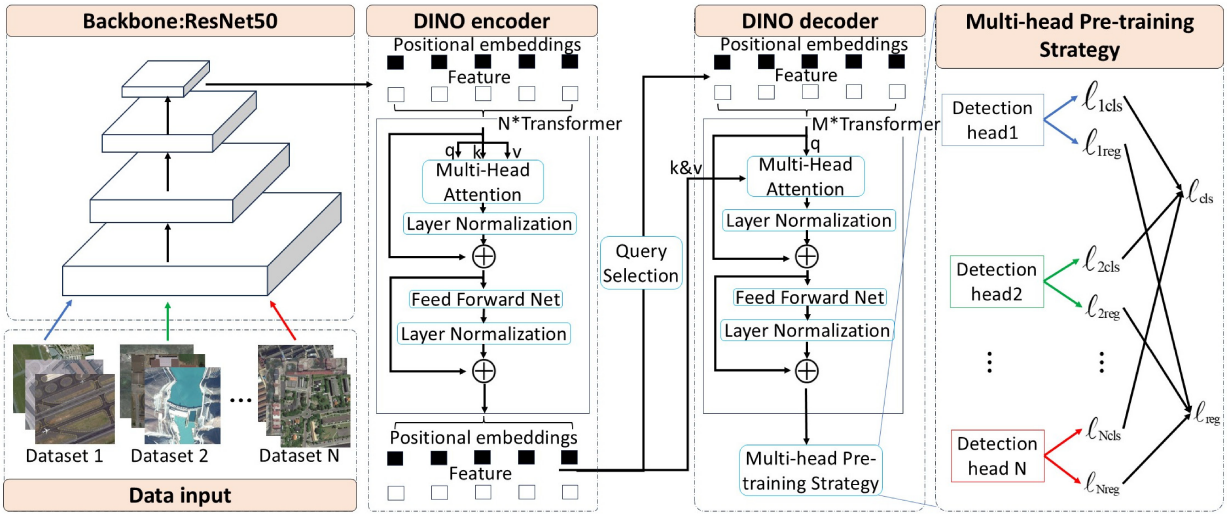

Fig. 3. Overall architecture of EON.

a multihead pretraining strategy to integrate data from diverse datasets. This approach expands the dataset scale, enabling joint training and facilitating object-level RS pretraining. Using this method, the constructed EON was pretrained on different large-scale RSOD datasets, to improve model performance on RSOD task.

As shown in Fig. 3, the EON architecture is based on detection transformer with improved denoising anchor boxes (DINO) [63], which consists of ResNet50 [64] backbone, transformer encoder, and transformer decoder. Considering the category differences across each object detection datasets, the multihead pretraining strategy assigns each dataset an independent detection head with output channels corresponding to the number of categories. During training, distinct minibatches are sampled from different datasets to form a composite large batch, which is fed into the backbone network and subsequently processed through the transformer encoder and decoder. Assuming a large batch size of B, small batch sizes are $B_i(i = 1, 2, 3, \ldots, N)$, then $B = \sum B_i(i = 1, 2, 3, \ldots, N)$. Assume that the size of the different datasets is L_i, B_i can be estimated as

$$B_i = \frac{L_i}{\sum_{j=1}^{3} L_j} \times B, \quad i = 1, 2, 3, \ldots, N. \tag{1}$$

The different large-scale object detection datasets are passed through the backbone and into the transformer encoder, where inference is conducted via the transformer decoder, outputting the corresponding classification losses and regression losses. We compute the classification loss ℓ_{cls} and regression loss ℓ_{reg} for each dataset as

$$\ell_{\text{cls}} = \sum w_i \times \ell_{i\text{cls}}, \quad (i = 1, 2, 3, \ldots, N) \tag{2}$$

$$\ell_{\text{reg}} = \sum w_i \times \ell_{i\text{reg}}, \quad (i = 1, 2, 3, \ldots, N). \tag{3}$$

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The weight w_i is proportional to the size of each dataset, and the final total loss is computed as the weighted sum of the classification and regression losses, represented as

$$\mathcal{L} = \ell_{\text{cls}} + \ell_{\text{reg}} = \sum_{i=1}^{N} \left(w_i \times \left(\ell_{i\text{cls}} + \ell_{i\text{reg}} \right) \right), \quad w_i = \frac{L_i}{\sum_{j=1}^{N} L_j}. \tag{4}$$

B. Scenario–Object Relationship Module

SORM consists of backbone, feature pyramid network, and GCFM. The backbone uses ResNet50, initialized with the pretrained backbone from EON. Meanwhile, the feature pyramid network is used to learn multiscale features of RS objects, thereby improving object detection capabilities.

In the real world, the distribution of geospatial objects follows certain patterns, and the scenes where similar objects appear often exhibit high similarity. Thus, effective module of the relationship between objects and their surrounding scenes becomes a critical factor in improving detection accuracy. Inspired by this, the GCFM is designed to extract feature embeddings of the whole scene from the deepest layer of the backbone and simultaneously uses the multiscale features from the FPN to model object embeddings. By interacting between these two embeddings, a scenario–object relationship map is generated, enabling the utilization of scene information to enhance the discrimination of object features and reduce background interference.

Given a high spatial resolution image $I \in R^{N\times 3\times H\times W}$, where N represents the number of batch size. H and W are the height and the width of the image, respectively. Within the GCFM, the feature $G \in R^{N\times 2048\times (H/32)\times (W/32)}$ from the deepest layer of the backbone is used to extract potential scenario information, generating a global feature vector Gen$h \in R^{N\times 256\times 1\times 1}$. The extraction of this global feature is achieved through global average pooling followed by a 1×1 convolution, ensuring effective expression of global information. Then, the multiscale features F of the object embeddings can be obtained through the FPN, which has four layers. The global feature vector Gen$h$ interacts to generate the scenario–object relationship map.

For the feature $F_i \in R^{N\times 256\times (H/2^{i+1})\times (W/2^{i+1})} (i = 1, 2, 3, 4)$, a 1×1 convolutional transform is used to project the input features F_i into a semantic space identical to the global features Gen$h$ to obtain Fen$h_i \in R^{N\times 256\times (H/2^{i+1})\times (W/2^{i+1})}$. Next, the scene relationship feature $R_i \in R^{N\times 256\times (H/2^{i+1})\times (W/2^{i+1})}$ is generated by calculating the similarity between the global feature Gen$h$ and the projected feature Fenh_i

$$R_i = \text{Fen}h_i \otimes \text{Gen}h. \tag{5}$$

The generated scene relationship feature R_i is normalized using the sigmoid activation function to enhance nonlinear expressive capability and is then multiplied by F_i to obtain the output feature $Z_i \in R^{N\times 256\times (H/2^{i+1})\times (W/2^{i+1})}$. And the final features $Z \in R^{N\times 256\times (H/32)\times (W/32)}$ are obtained through progressive downsampling and residual connection

$$Z_i = \frac{1}{1+\exp(-R_i)} \cdot F_i, \quad Z = \sum_{i=1}^{4} \text{Downsample}^{i-1}(Z_i). \tag{6}$$

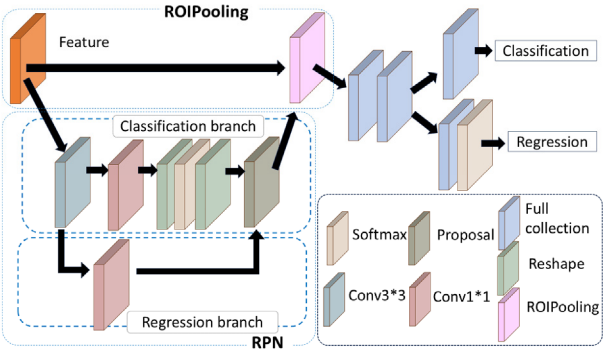

Fig. 4. Overall architecture of the detection head.

With the design of the GCFM, the network can better focus on global information while suppressing responses to irrelevant background, achieving more robust object detection and segmentation in complex scenes.

C. Detection Head

As shown in Fig. 4, the detection head consists of two main components: the RPN and ROI Pooling.

The RPN is used to generate positive anchor boxes. The RPN takes feature maps of an arbitrary size as input and outputs positive anchor boxes. The input feature $Z \in R^{N\times 256\times (H/32)\times (W/32)}$ is processed using a 3×3 convolution to generate the anchor box feature $E \in R^{N\times 256\times (H/32)\times (W/32)}$, which are then split into regression and classification branches. In the regression branch, the anchor box feature E is processed using a 1×1 convolution, a reshape module, and a SoftMax layer to predict the offsets (t_x, t_y, t_w, t_h) for each predefined anchor box

$$\begin{aligned} x &= x_a + t_x \cdot w_a, \quad y = y_a + t_y \cdot h_a \\ w &= w_a \cdot \exp(t_w), \quad h = h_a \cdot \exp(t_h) \end{aligned} \tag{7}$$

where (x_a, y_a) represents the predefined center coordinates of the anchor box, while w_a and h_a are the predefined width and height of the anchor box, respectively. (x, y, w, h) are presented as the adjusted anchor boxes.

In the classification branch, the adjusted anchor box features E are processed using 1×1 convolution to obtain the confidence scores for all the anchor boxes. The adjusted anchor boxes with confidence scores greater than the threshold are selected as positive anchor boxes and then put into ROI Pooling.

ROI Pooling is used to transform the positive anchor boxes generated by the RPN into a fixed size, facilitating further classification and regression. With the input feature Z and positive anchor boxes from the RPN, ROI Pooling divides each positive anchor boxes into $k \times k$ grids ($k = 7$) and applies max pooling within each grid

$$F_{ij} = \max_{(x,y)\in \text{cell}_{ij}} F(x, y) \tag{8}$$

where F_{ij} represents the pooling result of the i, j grid unit, and $F(x, y)$ denotes the feature value at the position (x, y) on the feature map.

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After ROI Pooling, the fixed-size feature map is fed into the classification and regression modules. Using fully connected layers, more accurate anchor boxes are obtained by refining the confidence scores and adjusting the anchor box locations for each category.

D. Label Smooth Loss

In RSOD tasks, the increase in spatial resolution amplifies intraclass variance and reduces interclass variance, leading to a higher possibility of misclassification. To address this, traditional object detection models often use the cross-entropy loss function, focusing on maximizing interclass variance to optimize classification performance. The cross-entropy loss function aids the model in fitting the data by maximizing the predicted probability of the true class. Assuming the model outputs the unnormalized score (logits) z_i for each class through a fully connected layer, these scores are converted into a probability distribution using the Softmax function

$$p_i = \frac{\exp(z_i)}{\sum_j \exp(z_j)} \tag{9}$$

where p_i represents the predicted probability of the class i. In classification tasks, labels are typically stored in a one-hot encoded format. For example, when the true class label is i, the predicted probability distribution q_{ki} for a sample which is predicted to class k is defined as follows:

$$q_{ki} = \begin{cases} 1, & (k = i) \\ 0, & (k \neq i). \end{cases} \tag{10}$$

Based on this, the cross-entropy loss can be defined as

$$\mathcal{L}_{\mathrm{CE}} = -\sum_i q_i \log(p_i). \tag{11}$$

This formula indicates that the optimization of the model for classification tasks is achieved by minimizing the negative log-probability of the true class. To further analyze the properties of the cross-entropy loss, we derive its gradient. For the model output z_i, the gradient of the softmax function is

$$\frac{\partial p_i}{\partial z_j} = p_i(\delta_{ij} - p_j) \tag{12}$$

where δ_{ij} represents the Kronecker delta function, which equals 1 when $i = j$ and 0 otherwise. By applying the chain rule, the gradient of the cross-entropy loss is derived as

$$\frac{\partial \mathcal{L}_{\mathrm{CE}}}{\partial z_j} = p_j - q_j. \tag{13}$$

For the true class j, the gradient is $p_j - 1$, while for other classes $i \neq j$, the gradient is p_j. This implies that as the model's predicted probability for the true class approaches 1, the gradient approaches 0, causing the model's training to slow down gradually. Although the cross-entropy loss performs well in classification tasks within deep learning, it has a notable drawback, as it drives the model to produce an extreme distribution of predicted probabilities. This extreme probability distribution may make the model more susceptible to background noise, reducing its adaptability. Moreover, when the model is trained on datasets with concentrated object labels, it may overlook intraclass variance, resulting in reduced generalization capability for unseen objects. To address the issues with cross-entropy loss mentioned above, the label smoothing technique has been proposed. It introduces a smoothing term in label distribution, reducing intraclass variance and thereby preventing the model from overfitting. Specifically, label smoothing transforms the one-hot encoded object distribution q_i into a smoothed distribution q_i^{smooth}

$$q_i^{\mathrm{smooth}} = (1 - \epsilon) \cdot q_i + \epsilon / K \tag{14}$$

where K denotes the total number of classes, and ϵ is the smoothing coefficient. By adding the smoothing term, the object probability for the correct class is set to $1 - \epsilon$ instead of 1, while the probability for other classes is distributed as (ϵ/K). Using this approach, the model's loss function is modified as

$$\mathcal{L}_{\mathrm{LS}} = -\sum_i q_i^{\mathrm{smooth}} \log(p_i). \tag{15}$$

This approach reduces the model's excessive reliance on a single class while assigning some probability weight to other classes, helping to prevent overfitting. Substituting the smoothed label distribution into the cross-entropy loss formula

$$\mathcal{L}_{\mathrm{LS}} = -\sum_j \left[(1 - \epsilon) \cdot q_i + \epsilon / K\right] \log(p_i) \tag{16}$$

expand to get it

$$\mathcal{L}_{\mathrm{LS}} = (1 - \epsilon) \cdot \mathcal{L}_{\mathrm{CE}} + \epsilon \cdot H(u, p) \tag{17}$$

where $H(u, p) = -\sum_i (1/K) \log(p_i)$ represents the cross-entropy between the predicted distribution p and the uniform distribution u, with the probability of each class becoming $(1/K)$. By introducing a uniform distribution, label smoothing regularizes the training process to a certain extent, preventing the model from becoming overconfident in any particular class. Label smoothing not only assigns probability weight to each class, avoiding the model's excessive reliance on specific classes, but also reduces interclass variance. Furthermore, by distributing the smoothed labels, label smoothing makes representations for the same class more compact, thereby reducing the overfitting phenomenon between similar classes, improving the model's generalization ability, and ultimately enhancing the robustness of object detection models in complex environments.

IV. Experiments

A. Datasets

1) NWPU VHR-10: NWPU VHR10 [65] includes ten categories and was proposed by Northwestern Polytechnical University in Xi'an and contains 1172 images, each 400 × 400 pixels, with 3651 object instances. RS objects are divided into airplane (PL), SH, storage tank (ST), baseball diamond (BD), tennis court (TC), basketball court (BC), ground track field (GTF), harbor (HB), bridge (BR), and vehicle (V). We use 75% of the dataset as the training set and 25% as the test set.

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TABLE I
INFLUENCE OF DIFFERENT LARGE-SCALE RSOD DATASETS ON EON

DIOR	HRRSD	LEVIR	DOTA	HRSC mAP (%)
✓				85.2
	✓			86.2
		✓		85.3
			✓	85.8
✓	✓			86.3
✓		✓		86.5
✓			✓	86.3
	✓	✓		86.6
	✓		✓	86.1
		✓	✓	86.4
	✓	✓	✓	87.0
✓	✓		✓	87.1
✓		✓	✓	86.2
✓	✓	✓		87.6
✓	✓	✓	✓	**87.8**

2) DOTA: DOTA [66] is a multiscale optical RS dataset that includes images from different platforms including Google Earth, the GF-2, and JL-1 satellites. Each image size ranges from 800 × 800 to 4000 × 4000 pixels. The DOTA dataset contains 188 282 object instances, with 15 object types, including airplane (PL), SH, ST, BD, TC, BC, GTF, HB, BR, large vehicle (LV), small vehicle (SV), helicopter (HE), roundabout (RO), soccer ball field (SBF), and swimming pool (SP). The training set contains 1411 RS images, and the validation set includes 458 images with varying object scales. We crop all the images to 1024 × 1024 with a stride of 824.

3) HRSC2016: HRSC2016 [67] contains 1061 images from Google Earth, including 70 marine images (90 samples) and 991 sea–land images (2886 samples). The spatial resolution of images ranges from 0.4 to 2 m. The image sizes vary from 300 × 300 to 1500 × 900 pixels, with most being larger than 1000 × 600 pixels. Categories include aircraft carrier, military SH, civilian SH, and submarine.

4) DIOR: DIOR [8] is the largest optical RSOD dataset in terms of the number of images and object categories. The DIOR dataset contains 23 463 optical RS images and 192 472 object instances, covering 20 common object categories such as airplane, airport, baseball field, BC, BR, chimney, dam, expressway service area, expressway toll station, golf course, GTF, HB, overpass, SH, stadium, ST, TC, train station, vehicle, and windmill. Images collected from Google Earth in DIOR have the size of 800 × 800 pixels with spatial resolutions ranging from 0.5 to 30 m.

5) HRRSD: HRRSD [68] comprises 21 761 images obtained from Google Earth and Baidu Maps, with spatial resolutions ranging from 0.15 to 1.2 m. It includes 13 object categories and 55 740 object instances. The 13 categories are SHs, airplanes, baseball fields, BCs, BRs, intersections, ground tracks, HBs, parking lots, oil tanks, T-junctions, TCs, and vehicles.

6) LEVIR: LEVIR [69] consists of a large number of high-resolution images from Google Earth, with 21 952 images with size of 800 × 600 pixels and resolutions ranging from 0.2 to 1.0 m. It includes three object categories and 11 028 objects instances, including 4724 airplanes, 3025 SHs, and 3279 oil tanks. Each image contains an average of 0.5 objects. LEVIR covers most types of ground features in human living environments, such as urban areas, rural areas, mountainous regions, and oceans. Extreme terrestrial environments such as glaciers, deserts, and gobi deserts are not included in our dataset.

B. Implement Specifications

For comparison, we selected commonly used methods, including Faster R-CNN, Retina R-CNN, Grid R-CNN, Cascade R-CNN, FoveaBox, and DINO, which are implemented on the mmdetection project.

Our experimental setup used the PyTorch framework (version 1.6) on a Linux system. All the experiments were conducted on four NVIDIA RTX 4090 GPUs, each with 24 GB of memory. Model parameters were optimized using the stochastic gradient descent (SGD) algorithm with an initial learning rate of 0.01, and a cosine annealing algorithm was used for learning rate decay. The batch size per GPU was set to 4. In addition, the mean average precision (mAP) at a 0.5 IoU threshold was used to evaluate detection performance across all the methods.

C. Evaluation Indexes

We use average precision (AP) and mAP across all the categories as evaluation metrics. Typically, a higher mAP value indicates better detection performance.

We also report the result of comparative experiments in the metrics of mAP50:95, which averages the precision over IoU thresholds ranging from 0.5 to 0.95 in steps of 0.05. The results are reported in Appendix of supplementary files.

D. Ablation Experiment of Different Large Scale Object Detection Datasets on EON

To evaluate the effectiveness of different large-scale RSOD datasets for pretraining, we conducted experiments by pretraining the EON model on various combinations of four available large-scale object detection datasets. We adjusted the number of datasets in each combination, selecting between one and four datasets and fine-tuned the model on the HRSC2016 dataset using the Faster R-CNN framework. This allowed us to explore the impact of different dataset combinations on the performance of EON. As shown in Table I, increasing

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TABLE II
COMPARISON OF METHODS ON THE HRSC2016 DATASET (TEST PART). BOLD VALUES INDICATE THE HIGHEST VALUE IN EACH COLUMN

Method	backbone	mAP (%)	Aircraft Carrier	Military Ship	Civilian Ship	Submarine	FPS
Faster RCNN	r50+IMP	85.2	93.0	93.0	87.0	67.7	20.4
	r50+RSP	86.9	94.2	96.6	83.8	72.9	20.1
	r50+EON	87.6	93.6	95.9	84.4	76.6	20.0
CascadeRCNN	r50+IMP	86.5	92.2	94.1	83.7	75.9	21.9
	r50+RSP	86.8	94.0	94.8	81.6	75.9	22.3
	r50+EON	87.3	87.9	95.9	85.5	80.1	**23.0**
GridRCNN	r50+IMP	84.4	86.1	95.5	84.8	71.4	16.1
	r50+RSP	85.8	86.1	94.8	83.6	78.6	15.9
	r50+EON	86.8	92.8	95.4	85.0	74.1	16.0
Retina RCNN	r50+IMP	85.2	93.0	93.0	87.0	67.7	16.2
	r50+RSP	84.0	91.8	**99.2**	79.9	72.0	16.0
	r50+EON	86.2	94.2	92.3	83.2	75.2	16.2
FoveaBox	r50+IMP	84.1	90.7	93.5	77.9	74.1	18.8
	r50+RSP	84.8	90.0	93.1	82.6	73.3	19.3
	r50+EON	87.0	90.7	92.7	84.3	80.3	19.6
DINO	r50+IMP	87.0	89.2	96.1	**90.1**	72.6	10.1
	r50+RSP	86.7	91.7	97.2	84.2	73.7	9.7
	r50+EON	87.2	91.3	95.7	86.7	75.2	9.8
PSODNet	r50+EON	**89.6**	**95.1**	96.4	82.7	**84.2**	20.0

TABLE III
COMPARISON OF METHODS ON THE DOTA DATASET (TEST PART). BOLD VALUES INDICATE THE HIGHEST VALUE IN EACH COLUMN

Method	Backbone	mAP	SV	LV	PL	ST	SH	HB	GTF	SBF	TC	SP	BD	RO	BC	BR	HE	FPS
Faster RCNN	r50+IMP	72.1	47.9	77.5	91.6	50.8	59.2	89.6	73.1	63.8	92	80	75.1	61.2	72.2	61	87.2	24.52
	r50+RSP	72.2	47.5	77.1	91.6	52.3	59.3	90.6	73.4	64.7	92	79.6	76	61.8	68.3	61	87.2	23.74
	r50+EON	74.3	48.8	80.1	92.5	52.5	60.3	89.8	74.2	69.1	93.9	81	80.1	66	76.4	64	87.2	24.14
Cascade RCNN	r50+IMP	70.7	41.9	73.5	91.3	53	56.4	91.1	68.3	65.5	93.2	85.1	78.6	63.2	66.2	60	72.7	**28.24**
	r50+RSP	70.5	41.2	74.7	90.6	56.5	56.7	87.6	68	62.2	93.7	82.7	77.6	60.8	77.9	55.5	70.9	27.95
	r50+EON	71.6	41.7	74.4	91.6	57	57.1	91.5	72.1	66.2	92.4	**85.4**	76.3	65.3	71.4	61.1	70.3	28.14
Grid RCNN	r50+IMP	71.6	46.9	72.1	91.4	51.3	59.1	89.2	70.8	62.6	92.9	78.5	76.6	61.5	72.4	62.1	86.2	24.64
	r50+RSP	72.9	47.8	73.6	92	52.3	59.1	87.7	75.5	68.4	93.7	78.5	79.5	61.8	74.8	62.8	85.6	24.55
	r50+EON	72	46.6	72	91.4	51.5	58.3	90.4	73.6	65.5	91.9	78.9	78.6	61.2	71	63.2	86.3	24.64
Retina RCNN	r50+IMP	72.4	47.2	78.5	91.7	51.6	59.3	90.6	71.6	63.1	92	80.8	78	63.1	69.2	63	86.1	19.12
	r50+RSP	73.8	47.1	80.6	91.4	55.1	59.9	87.6	75.4	67.3	93	79.5	80.8	69.7	75.3	62.6	81.9	18.94
	r50+EON	72.4	46.8	78.4	91.7	53.5	59.2	91.4	73.4	65.3	91	78.2	76	62.4	69.4	63	86	19.21
FoveaBox	r50+IMP	72.6	51.3	77.1	92.4	60.7	58.6	91.7	63.7	62.7	93.7	82.5	75.6	62.3	72.1	57.5	86.3	24.92
	r50+RSP	72.7	50.5	77.7	93	60.7	58.7	90.9	65.8	62.2	93.7	80.7	77.5	63.7	72.2	57.5	85.4	24.82
	r50+EON	73.2	52.1	77.5	92.9	62.3	58.9	**92.9**	68	60.2	92.9	82.3	77.3	65.4	68.6	59.6	86.7	24.8
DINO	r50+IMP	73.9	68.2	84	92.5	71.5	87.6	87.3	61.8	59.6	92.1	74.5	70.7	54.6	69.8	53.3	81.3	12.04
	r50+RSP	74.8	**70.1**	**85.1**	**93.4**	71.6	**88**	89.6	69.2	63.7	93.8	73	75.2	53.3	74.6	57.8	63.4	11.82
	r50+EON	75.5	69.6	81.9	93.1	69.4	87.7	87.3	70.4	68.1	94.5	74.5	76.9	62.3	71.9	56.7	**68.4**	11.58
PSODNet	r50+EON	**76.9**	50.4	81.8	92.4	**74.5**	59.3	84.3	**84.5**	**78.0**	**95.5**	65.3	**87.2**	**78.9**	**86.8**	**69.2**	63.4	24.3

the number of pretraining datasets generally improves performance of EON in fine-tuning. The model pretrained on DIOR, HRRSD, LEVIR, and DOTA achieves the highest mAP of 87.8%, demonstrating the benefit of diverse and large-scale pretraining.

However, the combination of DIOR, HRRSD, and LEVIR also achieves second highest mAP of 87.6%, which is only 0.2% lower than using all the four datasets. Considering this difference, we ultimately selected DIOR, HRRSD, and LEVIR for EON pretraining which requires less data and reduces training cost.

E. Comparative Experiments and Analysis

This section analyzes the comparison experiments of PSODNet with other common detectors on HRSC2016, DOTA, and NWPU VHR-10. The results are shown in Tables II–IV.

1) Results on HRSC2016 Dataset: In our study, we evaluated the performance of various object detection methods on the HRSC dataset. We compared multiple detection methods with our method, including Faster R-CNN, Cascade R-CNN, Grid R-CNN, Retina R-CNN, FoveaBox, and DINO, each with different backbone configurations (such as r50 + IMP, r50 + RSP, and r50 + EON). As shown in Table II, EON achieved better results than IMP and RSP under the same object detection methods, and PSODNet demonstrates outstanding results across multiple categories. This improvement is mainly attributed to PSODNet's SORM, which leverages scene and object information. Combined with EON, which has been trained on sufficient RSOD data, the PSODNet's ability to accurately detect various objects in complex scenes is further improved.

For instance, in the aircraft carrier detection task, PSODNet achieved the mAP of 95.1%, while other methods generally yielded lower APs for the same category, with the second-best mAP being 94.2%, marking a 0.9% improvement. This result highlights the significant advantage of PSODNet in RSOD, especially in detecting large objects in complex scenes. In the submarine detection, PSODNet also led with the mAP of 84.2%, outperforming the next best method by 3.9%, which underscores the model's strong adaptability to small-sized objects. These results indicate that PSODNet excels in multicategory object detection, primarily due to its ability to

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TABLE IV
COMPARISON OF METHODS ON THE NWPU VHR10 DATASET (TEST PART). BOLD VALUES INDICATE THE HIGHEST VALUE IN EACH COLUMN

Method	backbone	mAP (%)	PL	SH	ST	BD	TC	BC	GTF	HB	BR	V	FPS
Faster RCNN	r50+IMP	88.3	99.9	73.6	84.3	97.9	89.9	80.2	**100.0**	91.9	77.6	87.9	19.9
	r50+RSP	90.6	100.0	75.2	90.6	99.4	89.0	**100.0**	**100.0**	85.1	78.4	87.8	20.2
	r50+EON	92.6	100.0	77.5	85.3	98.4	96.7	**100.0**	**100.0**	92.1	79.4	96.5	20.1
CascadeRCNN	r50+IMP	90.4	99.9	77.1	90.7	97.8	93.6	**100.0**	99.3	91.2	69.0	98.4	**21.5**
	r50+RSP	91.1	99.9	75.6	90.1	97.8	94.6	**100.0**	**100.0**	90.6	80.2	91.1	21.2
	r50+EON	92.2	**100.0**	77.6	80.7	95.1	97.0	**100.0**	99.3	90.8	74.6	96.0	21.2
GridRCNN	r50+IMP	91.3	**100.0**	76.2	89.1	97.6	97.6	**100.0**	**100.0**	91.9	66.8	93.7	16.0
	r50+RSP	91.4	**100.0**	73.7	**93.2**	**99.9**	97.6	**100.0**	**100.0**	90.1	66.4	93.3	16.0
	r50+EON	91.7	**100.0**	76.7	87.1	97.4	97.6	**100.0**	99.3	88.0	79.6	90.9	14.5
Retina RCNN	r50+IMP	91.3	**100.0**	79.0	83.8	98.7	83.2	**100.0**	**100.0**	89.6	66.3	**99.8**	16.1
	r50+RSP	92.0	**100.0**	73.2	90.9	99.6	97.6	96.7	**100.0**	81.2	78.6	93.8	16.1
	r50+EON	92.0	**100.0**	77.0	88.0	97.4	**98.7**	**100.0**	**100.0**	88.1	84.7	96.6	16.5
FoveaBox	r50+IMP	90.9	**100.0**	79.8	82.4	99.5	94.6	80.2	99.7	90.8	90.9	91.3	18.4
	r50+RSP	91.7	99.9	73.4	88.0	99.7	94.4	80.2	**100.0**	89.8	**100.0**	91.1	18.2
	r50+EON	92.1	**100.0**	80.1	83.2	98.8	86.8	**100.0**	**100.0**	95.4	83.5	93.5	17.8
DINO	r50+IMP	91.6	99.9	72.2	87.7	97.9	95.3	**100.0**	**100.0**	91.9	78.1	93.4	9.9
	r50+RSP	90.4	99.4	72.9	87.9	98.2	94.4	**100.0**	**100.0**	78.2	82.8	89.9	9.8
	r50+EON	92.4	**100.0**	75.6	86.9	98.0	96.2	**100.0**	**100.0**	93.9	80.6	93.3	9.4
PSODNet	r50+EON	**95.0**	**100.0**	**88.6**	91.7	99.6	93.6	96.7	**100.0**	**96.6**	84.8	98.7	19.7

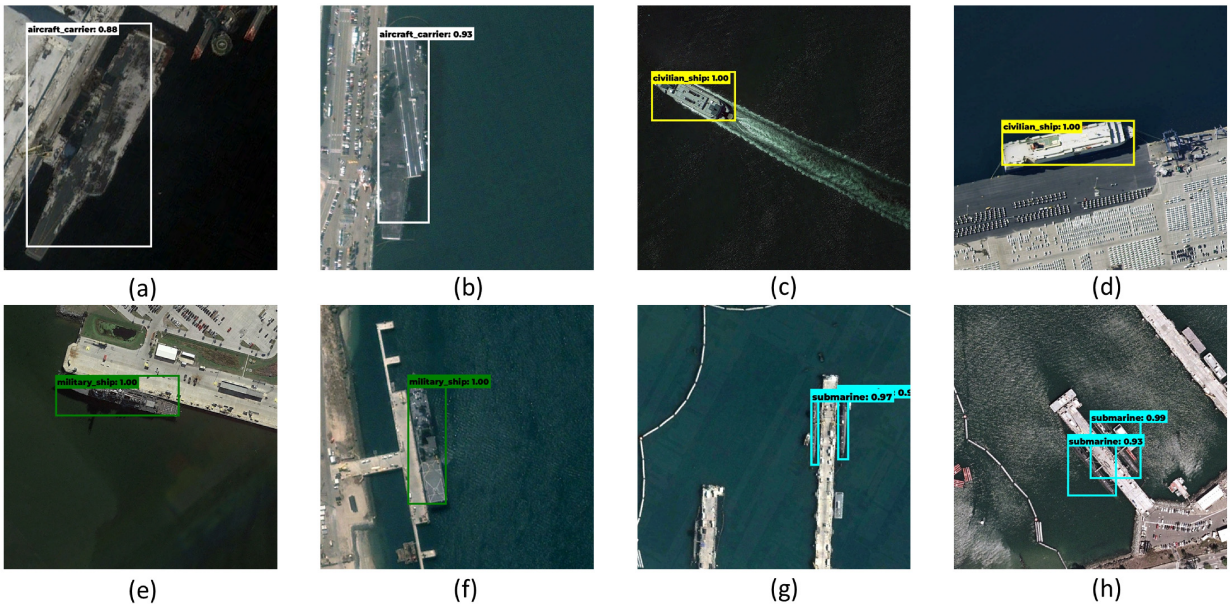

Fig. 5. Representative detection results of PSODNet on the HRSC2016 dataset. (a) and (b) Aircraft carrier. (c) and (d) Civilian SH. (e) and (f) Military SH. (g) and (h) Submarine.

capture and use scene information to enhance object feature detection when dealing with multiscale objects.

In Table II, Cascade RCNN (r50 + IMP) achieves the highest FPS of 23.0, indicating superior inference efficiency. PSODNet achieves 20.0 FPS, maintaining a good balance between speed and accuracy. In contrast, DINO variants exhibit lower speeds (as low as 9.7 FPS), reflecting higher computational complexity. The detection results are shown in Fig. 5.

2) Results on DOTA Dataset: We report the mAP of different methods on the test part of the DOTA dataset. Table III represents the mAP (%) score for each category. Among all the tested models, PSODNet combined with the r50 + EON backbone achieved the highest overall mAP, reaching 76.9%. Specifically, this combination achieved the best performance in eight out of the 15 categories, including GTF with the mAP of 84.5%, ST with the mAP of 74.5%, SBF with the mAP of 78.0%, TC with the mAP of 95.5%, BD with the mAP of 87.2%, RO with the mAP of 78.9%, BC with the mAP of 86.8%, and BR with the mAP of 69.2%. However, we observed that it performed poorly in detecting SV, achieving only 50.4% mAP in this category. This may be attributed to the high density of SVs and minimal feature differences between them, making differentiation challenging and leading to lower performance in this category.

As shown in Table III, Cascade RCNN (r50 + IMP) reaches the highest FPS of 28.2. PSODNet achieves 24.3 FPS, offering competitive efficiency while maintaining a high accuracy. The detection results are shown in Fig. 6.

3) Results on NWPU VHR10 Dataset: We report the mAP of different methods on the test part of the NWPU VHR10 dataset. Table IV lists the results for each comparison method. Each column represents the mAP (%) score for each category.

The mAP results show that PSODNet, combined with the r50 + EON backbone, achieved the highest mAP score of

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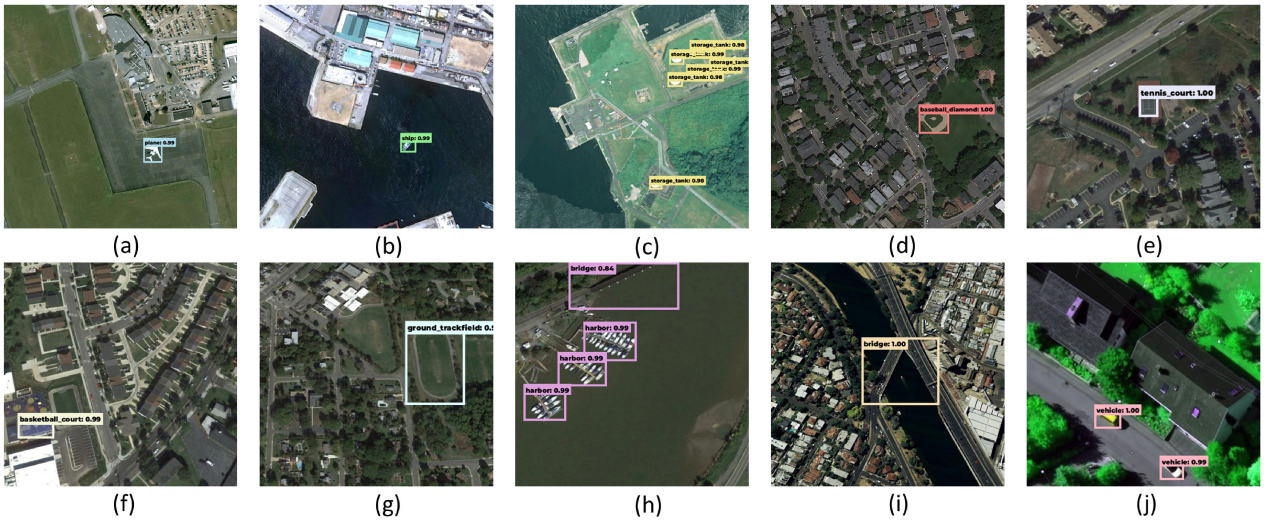

Fig. 6. Visualization of PSODNet detection results on the DOTA dataset. (a) Airplane. (b) SH. (c) ST. (d) BD. (e) TC. (f) BC. (g) GTF. (h) HB and BR. (i) BR. (j) Vehicle.

TABLE V
ABLATION EXPERIMENT

Method	EON	SORM	Label Smooth Loss	HRSC mAP (%)	DOTA mAP (%)	VHR10 mAP (%)
(a)Baseline				85.2	72.1	88.3
(b)Baseline+EON	✓			87.6	74.3	92.6
(c)Baseline+EON+ Label Smooth Loss	✓		✓	89.1	75.6	94.2
(d)Baseline+EON+SORM	✓	✓		88.3	76.0	93.8
(e)PSODNet	✓	✓	✓	**89.6**	**76.9**	**95.0**

95.0%, significantly outperforming other methods. PSODNet excelled across multiple categories, achieving perfect detection in the PL and GTF categories with the mAP of 100.0%. It also showed strong performance in detecting large objects, such as SHs (88.6%) and HBs (96.6%). Furthermore, PSODNet demonstrated high detection accuracy for complex-background objects such as vehicles (V, 98.7%), STs (91.7%), and BD (99.6%), achieving near-optimal results in these categories.

In Table IV, PSODNet achieves 19.7 FPS, slightly lower than Faster RCNN and Cascade RCNN (up to 21.5 FPS). PSODNet achieves the highest accuracy while maintaining high runtime, making a balance in terms of both performance and speed. The detection results are shown in Fig. 7.

F. Visualization of Model Detection Results in Complex Conditions

To further evaluate PSODNet's detection performance in complex environments, we selected three different scenarios, as shown in Fig. 8. In cases of shadow occlusion [Fig. 8(a)] and multiscale effects [Fig. 8(b)], PSODNet effectively reduces missed detections compared with other methods, thus improving detection accuracy. When similar objects are in close proximity [Fig. 8(c)], PSODNet captures subtle boundaries between objects, achieving precise differentiation and reducing false detections.

G. Ablation Analysis and Discussion

To evaluate the contribution of each module in PSODNet, we conducted an ablation study to analyze their importance. The results are shown in Table V. The baseline model consists only of a ResNet50 backbone and a detection head, with IMP as the pretrained model. Adding EON to the baseline [Table V(b)] improved mAP by 2.4% on HRSC2016, 2.1% on DOTA, and 4.3% on NWPU VHR10, confirming the importance of RS-specific pretraining for object detection in RS images. Further inclusion of Label Smoothing Loss [Table V(c)] and SORM [Table V(d)] yielded additional improvements of up to 1.6% and 1.7% in mAP, respectively, indicating that learning scene information and smoothing label distribution are crucial for accurate object detection in complex backgrounds. When both label smoothing loss and SORM were applied together [Table V(e)], mAP increased further to 89.6% on HRSC2016, 76.9% on DOTA, and 95.0% on NWPU VHR10.

H. Visual Interpretation of the Scenario-Object Relationship Map

The SORM, integrated with geoscientific knowledge, provides strong visual interpretability. Fig. 9 visualizes scenario–object relationships at different pyramid levels, where each pixel represents the strength of the relationship between a potential geospatial scene and the pixel itself. Three typical scenes are depicted: HB, airport, and athletic field.

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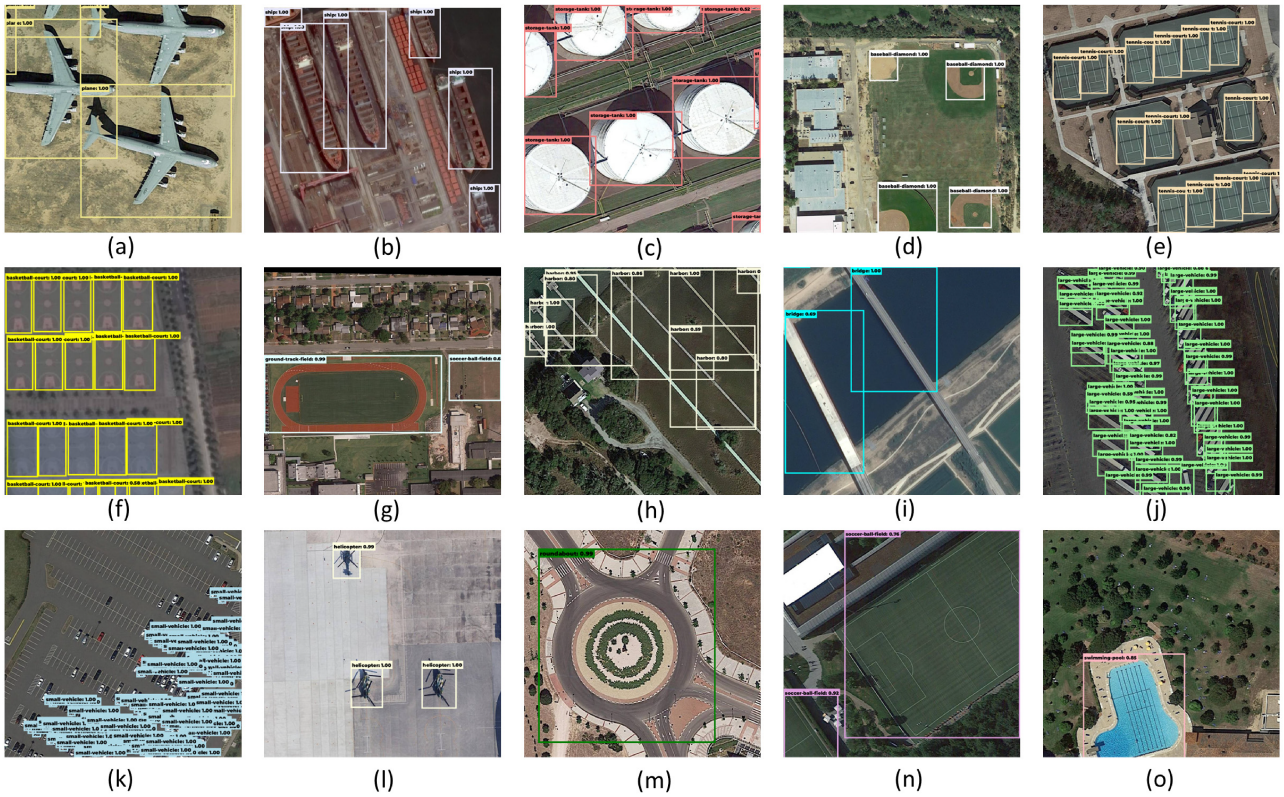

Fig. 7. Visualization of PSODNet detection results on the NWPU VHR-10 dataset. (a) Airplane. (b) SH. (c) ST. (d) BD. (e) TC. (f) BC. (g) GTF. (h) HB and BR. (i) BR. (j) LV. (k) SV. (l) HE. (m) RO. (n) Soccer field. (o) SP.

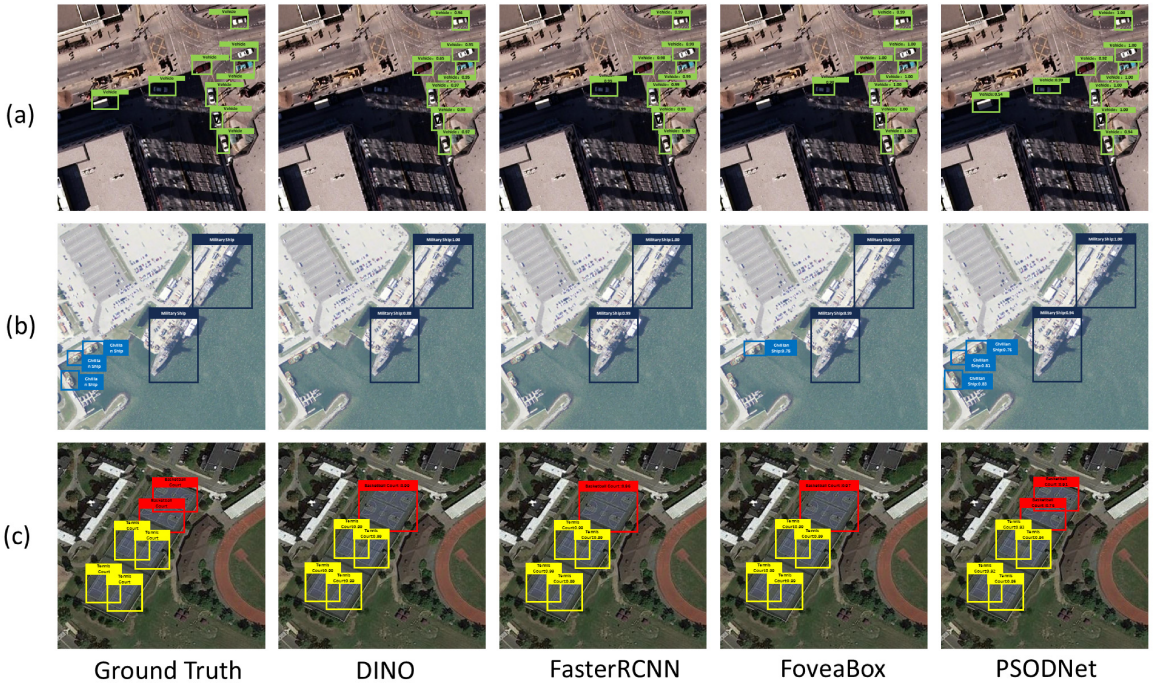

Fig. 8. Comparison of PSODNet detection results to other methods in complex scenes. (a) Shadow occlusion. (b) Multiscale objects. (c) Similar objects in close proximity.

We observe that each scene focuses on distinct objects that characterize the scene. For example, the HB scene mainly highlights SHs and docks, while the airport focuses primarily on airplanes and their surroundings. These relationship maps further illustrate the connection between geospatial scenes and multiscale objects.

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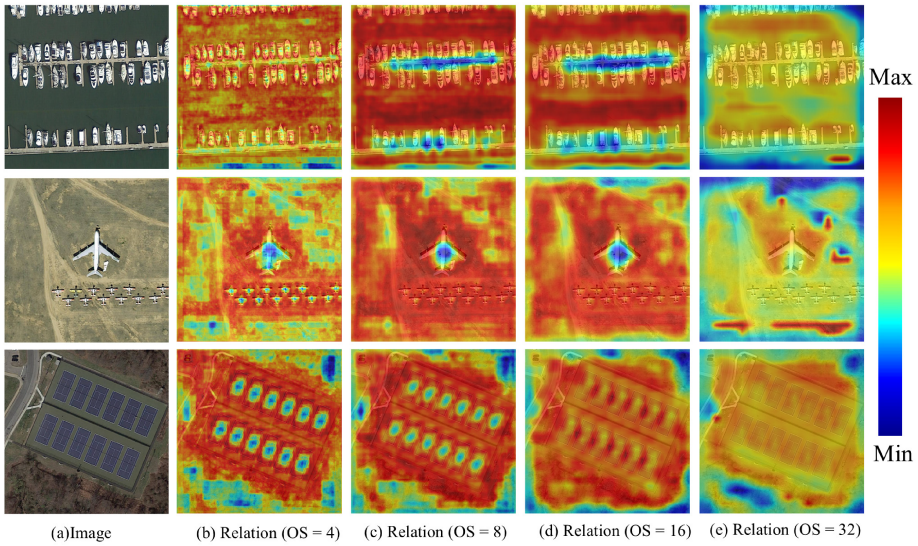

Fig. 9. Visualization of scenario–object relationship maps at different pyramid levels. (a) Original image. (b)–(e) Scenario–object relationship maps across different pyramid levels. OS represents the "output stride" as defined in the FPN. For visualization, we resized these relationship maps to match the size of the corresponding image. Legend: Scene 1 (HB), Scene 2 (airplane), and Scene 3 (TC), in sequence.

In high spatial resolution maps (OS = 4), attention is primarily directed to the objects themselves, such as the centers of airplanes and TCs. As the receptive field expands, the relationship maps increasingly capture scene-specific features, such as docks in HB scenes. This demonstrates that GCFM can effectively model the relationships between scenes and objects, using these relationships to constrain the solution space for object recognition. In addition, we observe that the model learns scenario–object relationships at multiple scales across different receptive fields, enriching the diversity of these relationships.

V. Conclusion

This article introduces a novel pretrained scene-aware object detection network aimed at enhancing the detection performance of multiscale objects in RS, particularly for detecting objects within complex backgrounds and under shadow interference. PSODNet effectively integrates scenario–object information and multiscale feature representation through an innovative design.

First, an EON replaces the traditional scene-level pretraining with object-level pretraining to address the domain gap and uses a multihead pretraining strategy to further improve the model's generalization ability. Second, the SORM is proposed to construct a scenario–object relationship map that constrains the solution space of object detection, enhancing detection performance in complex scenes. Finally, label smoothing loss is introduced to reduce intraclass variance, thereby preventing overfitting and improving the model's generalization.

Through a series of comparative experiments with various detection methods under identical dataset settings, it is found that PSODNet performs well in object detection across various complex scenarios. Feature visualization results further illustrate the connection between geospatial scenes and multiscale objects, which enhances the model's adaptability in various complex environments. The design of PSODNet offers new perspectives on addressing challenges including enhancing generalization, reducing background interference, and mitigating overfitting, which show potential in real-world uses including vehicle detection, burned area detection, and landslip detection. For instance, PSODNet can effectively detect vehicles in intricate traffic situations during peak hours, helping traffic authorities with management and control.

However, we acknowledge that PSODNet has certain limitations in handling ultrasmall objects in high-resolution images, which we aim to optimize in future research. Moreover, PSODNet has not been designed to address other challenges of ultrahigh-resolution imagery, including large data volume and complex object details. In future work, we will improve PSODNet to better adapt to ultrahigh-resolution imagery by enhancing ability to capture subtle differences in objects for complex object details and optimizing memory-efficient processing for large data volumes.

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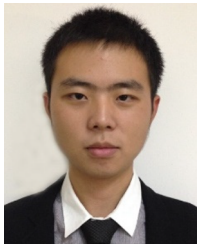

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